This is your fifth and final homework assignment. I strongly encourage you to typeset it using the LaTeX template provided. It is due by Friday, March 13th. After class on Wednesday I will add a question on Pagerank.

The following questions are on constrained optimization.

(1) The non-negative least squares problem is:

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathbb{R}^d:\ \mathbf{x}\geq 0}{\arg\min}\ \|A\mathbf{x}-\mathbf{b}\|_2^2$$

In this question we will discuss using projected gradient descent (PGD) for this problem. Note that we can write this as:

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathcal{X}}{\arg\min}\, f(\mathbf{x}) \quad \text{where } \mathcal{X} = \left\{\mathbb{R}^d :\ \mathbf{x}\geq 0\right\} \text{ and } f(\mathbf{x}) = \|A\mathbf{x}-\mathbf{b}\|_2^2$$

Recall that PGD is defined by the iteration $\mathbf{x}_{k+1} = \mathcal{P}_{\mathcal{X}}\left(\mathbf{x}_k - \alpha\nabla f(\mathbf{x}_k)\right)$. For this particular region $\mathcal{X}$, the projection is defined component-wise as:

$$\left[\mathcal{P}_{\mathcal{X}}(\mathbf{v})\right]_i = \begin{cases} v_i & \text{if } v_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Suppose that:

$$A = \begin{bmatrix} 0.5377 & -0.8622 \\ 1.8339 & 0.3188 \\ -2.2588 & 1.3077 \end{bmatrix} \quad \text{and } \mathbf{b} = \begin{bmatrix} -0.4336 \\ 0.3426 \\ 0.5163 \end{bmatrix}$$

Given $\mathbf{x}_0 = \begin{bmatrix} 0 & 0 \end{bmatrix}^\top$ use PGD to find $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$. You may do this by hand, or you may use a computer. Take the step size, α, to be 0.01. *Recall that in a previous homework assignment we established that* $\nabla f(\mathbf{x}) = 2A^\top(A\mathbf{x}-\mathbf{b})$.

(2) Consider the constrained least squares problem:

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathbb{R}^d:\ \|\mathbf{x}\|_2\leq 1}{\arg\min}\ \|A\mathbf{x}-\mathbf{b}\|_2^2$$

The constraint $\|\mathbf{x}\|_2 \leq 1$ serves to ensure that the entries in the solution are not too large. Instead of solving this directly, one can consider the *penalty method* to approximately solve the problem. This means solving:

$$\mathbf{x}^\# = \underset{\mathbf{x}\in\mathbb{R}^d}{\arg\min}\, f(\mathbf{x}) = \|A\mathbf{x}-\mathbf{b}\|_2^2 + \gamma\|\mathbf{x}\|_2^2$$

This problem is also called *ridge regression*. Let's also define:

$$\mathbf{x}^\dagger = \underset{\mathbf{x}\in\mathbb{R}^d}{\arg\min}\, \|A\mathbf{x}-\mathbf{b}\|_2^2$$

(a) As $\gamma \to 0$ do you expect $\mathbf{x}^\# \to \mathbf{x}^*$ or $\mathbf{x}^\# \to \mathbf{x}^\dagger$? Explain your answer.
(b) For intermediate values of γ, do you expect $\mathbf{x}^\# \approx \mathbf{x}^\gamma$ or $\mathbf{x}^\# \approx \mathbf{x}^\dagger$? Explain your answer?
(c) As $\gamma \to \infty$ what will happen to $\mathbf{x}^\#$? Explain your answer.

The following questions are on dynamic programming.

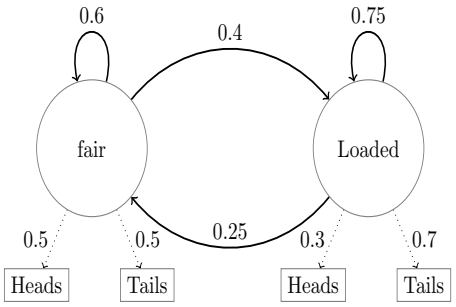
(3) Consider the following variant of the knapsack problem. Suppose that the Math 118 midterm consists of 5 problems, worth 5, 10, 20, 25, 30 and 35 points respectively. However, the test is only out of 50 points. You may attempt as many of the problems as you like, and your instructor will grade all of them and choose the subset of problems whose totals add up to less than or equal to 50 giving you the best possible score. Suppose you attempt all five questions. In what follows, p_i represents the points achieved on the i-th question, while t_i represents the total points available on the i-th question.

$$p_1 = 4, t_1 = 5 \quad p_2 = 3, t_2 = 10 \quad p_3 = 17, t_3 = 20 \quad p_4 = 20, t_4 = 25 \quad p_5 = 30, t_5 = 35$$

Use dynamic programming to determine the optimal subset of questions, and your maximum achievable score. *(This is essentially the knapsack problem with $B = 50$, $s_i = t_i$ and $v_i = p_i$. Hint: Consider increasing the size of the "knapsack" in increments of 5).* Your answer should include the dynamic programming table you construct as part of the algorithm.

(4) Use dynamic programming to find the longest common subsequence between $\mathbf{X} = GGATCGT$ and $\mathbf{Y} = CGAGCTT$. Your answer should include the dynamic programming table you construct as part of the algorithm.

(5) Consider the dishonest casino problem modeled by the Hidden Markov Model (HMM) given below. Suppose we observe the sequence of emissions $\mathbf{Y}_6 = HTHTTT$ *(this is this same sequence we worked*



with in class). Use Viterbi's algorithm to find the most probable sequence of hidden states. (Here, the hidden states are of course "F" and "L"). Assume an initial distribution of $\mathbb{P}[x_1 = F] = 1$ and $\mathbb{P}[x_1 = L] = 0$. Your answer should include the dynamic programming table you construct as part of the algorithm.

The following questions, still to be added, are on PageRank.

(6) In this question you will verify some things that were claimed, but not proven, in Lecture 22. The notation is as used in those slides.

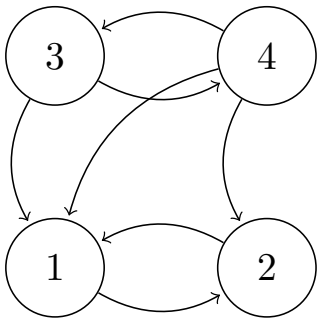
(a) Verify that $P_{ij} = \dfrac{(A^\top)_{ij}}{d_j^{\text{out}}}$

(b) Check that if P is column stochastic, then so is $\tilde{P}$. *Remark: there was a minor typo in the definition of $\tilde{P}$ given in class. It should be:*

$$\tilde{P} = \alpha P + (1-\alpha)\frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

Note the factor of $\frac{1}{n}$. Without this, $\tilde{P}$ will not be column-stochastic.

(7) Consider the directed graph given below:



(a) Write down the adjacency matrix A for this directed graph. Is A reducible or irreducible?

(b) Compute the matrix $\tilde{P}$ we defined in class, for $\alpha = 0.95$. *And use the definition of $\tilde{P}$ given in (6.b)*

(c) Implement the power method, using MATLAB or Python, and use it to compute the leading eigenvector, $\mathbf{v}_1$, of $\tilde{P}$. *You may want to play with different numbers of iteration of the power method, as well as different initial $\mathbf{v}$. You should also compare the approximation to $\mathbf{v}_1$ you get from power method with the true leading eigenvector, obtainable using the* `eig` *command in MATLAB*

(d) Using $\mathbf{v}_1$, rank the 4 vertices of this graph.